



Testplan for BB Fieldbus



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Author: Rieping, Christian

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 **Meier, Alexander**

read and reviewed

Signed

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 **Ruess, Bastian**

reviewed and ok

Signed

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 **Mouldi, Hajer**

reviewed and ok

Signed

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 **Zribi, Jamel**

Reviewed and ok

Signed

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 **John Wilkinson**

reviewed and ok

Signed

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 **Häfele, Philipp**

Reviewed and ok

Signed

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 **Schwarz, Ralph**

reviewed with verdict ok

Signed

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 **Schwarz, Ralph**

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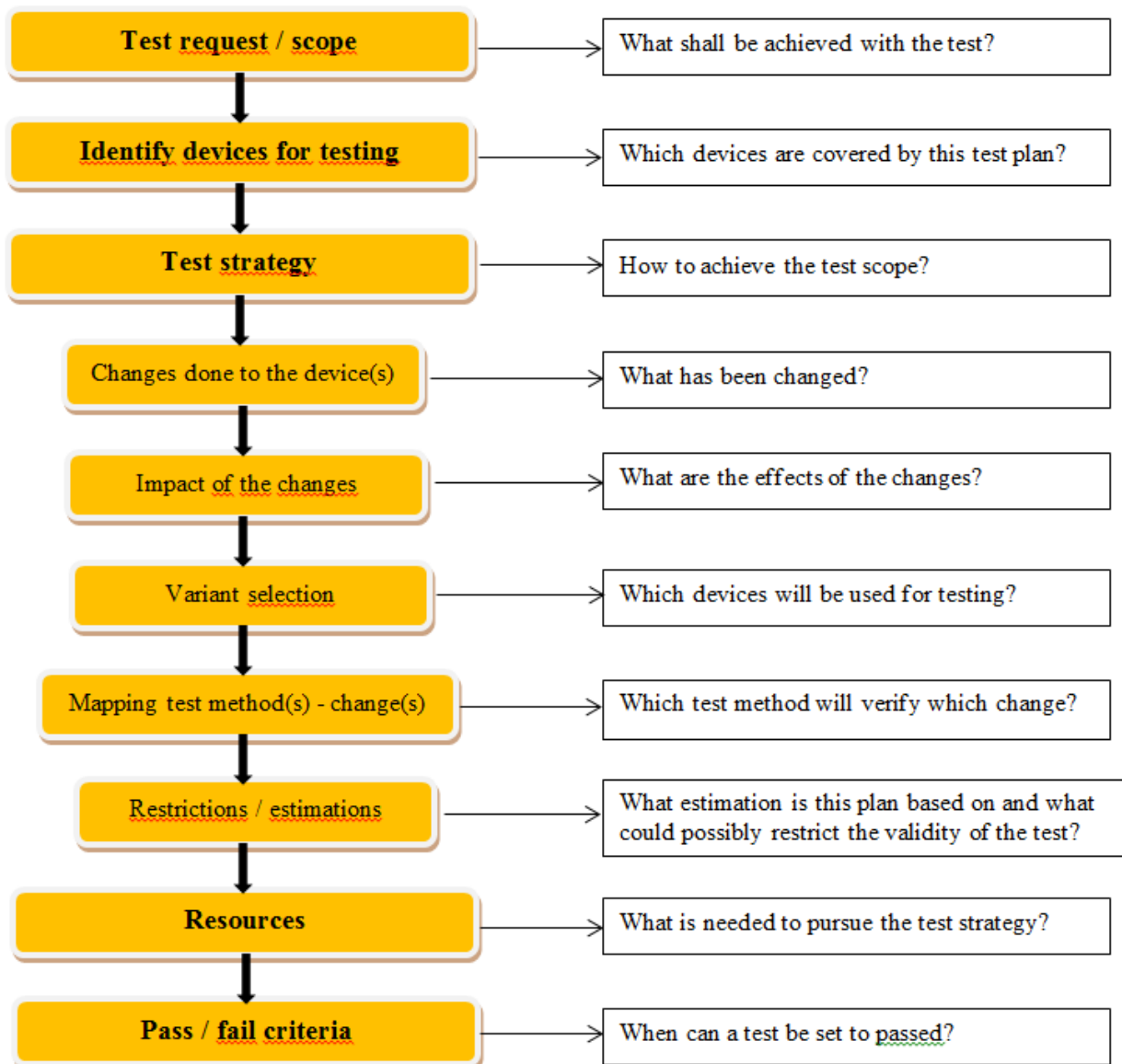
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Type	Chapter	Description
e,a,m	all	Added Ethernet/IP as planned Test. Marked in yellow

V01.01.001

Type	Chapter	Description
e,a,m	all	Testplan modified to fit BB-Fieldbus V1.1
m	all	reworking testplan based on review comments

1 General explanation and tips regarding the test plan





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2 Scope

This document describes the planned test activities for project "153469 - BB Fieldbus".

It defines the objectives and scope of the verification activities.

The test plan is the plan for the procedure and does not describe details of the test activities that are described in a separate test definition. This test plan does not contain any results, these are documented in a separate test report.

2.1 Applicable documents

R ef	Document name	Date	Version
[1]	Requirements -  VHIP BB Fieldbus		
[2]	SVN Folder for Profinet Tools		
[3]	SVN Folder for EthernetIP Tools		
[4]	SVN Folder for EtherCAT Tools		
[5]	SVN Folder for ModbusTCP Tools		
[6]	SVN SystemTestFolder		

2.2 Abbreviations

Abbreviation	Meaning
BB	Building Block
CCB	Change Control Board
DCP	Discovery and Configuration Protocol
DUT	Device Under Test
EMC	Electro-Magnetic Compatibility
GSDML	General Station Description Markup Language
GSD	General Station Description (file)
HW	Hardware
IO	Input/Output
IOP	Interoperability
ME	Mechanics
N/A	Not Applicable
OS	Operating System
PLC	Programmable Logic Controller
PN	PROFINET
QM	Quality Management
SVN	SubVersion
SW	Software



Abbreviation	Meaning
PM	Project Manager
tbd	To Be Defined
Ref.App.	Reference Application; which is a demonstration of some of the functionalities of the BB-Fieldbus, as defined in Requirement of BB-Fieldbus [1]



3 Scope of the test activities

The test plan describes the measures that are necessary for the validation and verification of the product (or its components). The test plan defines what needs to be tested and why we want to test it.

3.1 Test mission

When creating the test plan and the successive test specifications, the following character is taken into account development activity:

	Corrective action / component discontinuation
X	Custom change
X	Small software release with (few new features) [For Ref. App. Profinet and Ethernet/IP]
	Changes to OS components (startups, bootloaders)
X	New development [EtherCat and Modbus Ref. App.]
X	Further development [BB-Fieldbus in general]

3.2 Quality targets

When creating the test plan and the consecutive test specifications the following quality targets are to be taken into account:

	Performance
X	Function
	Power consumption
	Compatibility
X	Documentation
X	Interoperability

Reasons for not selected targets:

- **Performance:** Was not chosen due to the reference application is only a very small implementation and therefore the performance does not represent the expectable performance of a product implementation.
- **Power consumption:** The BB-Fieldbus is the ifm Fieldbus API and not the circuit on the EVAL-Board. Products will most likely have different circuits therefore there was no focus in reducing the power consumption.
- **Compatibility:** Compatibility to BB-Fieldbus V1.0 was not a focus because there are no users yet, also we wanted to update the stacks / added features.

3.3 Development phases according V-model

Test level verification	Product	HW	SW	ME
Type approval (versus requirements)				
System test / -integration (versus system specification)	X		X	
Module test (versus design / architecture)				
Implementation / Development test				



Test level verification	Product	HW	SW	ME
Please mark: X = Part of test plan				

4 Test object

The Building Block Fieldbus is a Stackintegration with an ifm specific Interface. The only functionalities which are testable are the ones which are used inside of an application, the so called reference application. There not every functionality is included, since the reference application shall provide an easy to use, fast entry in the "fieldbus world". Therefore the test object will be the reference application with the respective linked requirements for the implemented functionality.

4.1 New devices/components related to this project

To the already existing Ref. App. for PN and EIP, a reference application for ECAT and Modbus is developed which shall now be tested, additional to stack update and small changes in PN and EIP. Compared to BB-Fieldbus V1.0 it will now be tested on VHIP 3 Core EVAL Board with plugged BB-Fieldbus EVAL Board, therefore the PN and EIP Tests will be a full test to validate the behavior on the VHIP3 platform.

Consisting of the following components:

- The HW layout of the BB-Fieldbus (in form of an EVAL Board), which contains a Hilscher netX90 with PN/EIP/ECAT/Modbus stack.
The EVAL Board of the BB-Fieldbus is plugged on top of an EVAL Boards VHIP3 STM32F777.
- ifm fieldbus API with documentation for the later integrated application.
- Reference application for each fieldbus which shows a possible usage of the ifm fieldbus API

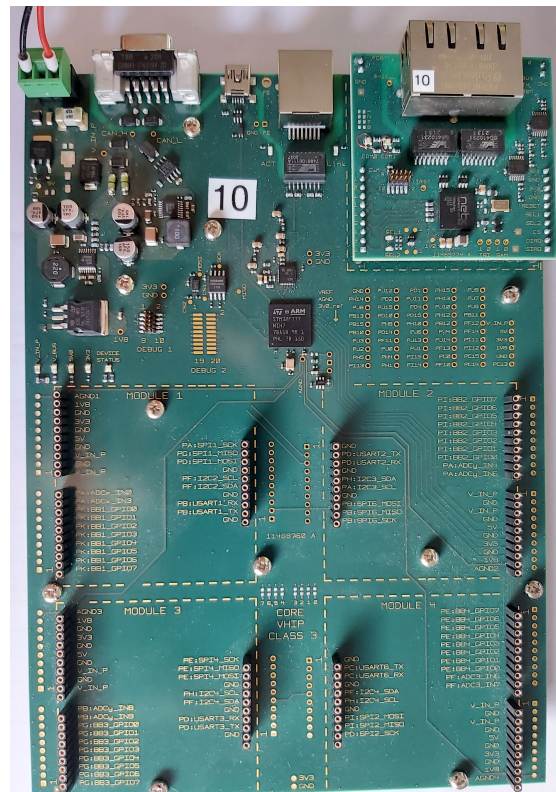
Definition of functionalities are placed here:

[Profinet Ref.App.](#)

[Ethernet/IP Ref.App.](#)

[EtherCAT Ref.App.](#)

[Modbus Ref.App.](#)



4.2 Firmware releases

Releases for BB-Fieldbus will be uploaded to [Nexus](#)



5 Test strategy

5.1 Change and impact analysis

5.1.1 Changes to the DUT

	Assembly option
X	Changed schematics [EVAL Board for BB-Fieldbus]
	Functional housing change (or change of functional parts)
	SW change of start-up components (Boot loader, Init-routines)
X	SW change of functional components [Small changes for PN and EIP]
	SW change of maintenance components (on PC)
X	New product: Ref. App. for EtherCAT
X	New product: Ref. App. for Modbus

5.1.2 Open Change Requests and Defects from V1.0

N o.	Type of change	Change of behaviour	Possible impact	Type of test	Test-No.:
1	☐ VHIP-3210 - RefApp Profinet: PN-Tester skipped one Testcase because not testable due to Setup	-	-	Conformance Test will be repeated for V1.1; Verify if Setup is the issue.	-
2	☐ VHIP-3928 - RefApp Profinet: Deactivate of second Ethernet Port not working	-	-	Was found in BB-Fieldbus Version 1.0 with VHIP4 Board. Will be tested again with VHIP3 HW	-
3	☐ VHIP-3687 - RefApp Profinet: Online View in TIA does not show the correct state of the Subslot	Not yet clear if we can do something, or it is Siemens TIA related.	To be defined as soon as change (if necessary) is known	Ongoing, poll the status @B.Ruess	-
4	☒ VHIP-3744 - VHIP4 RefApp EIP: Read internal state of Module is wrong	Fixed in Hilscher Stack PSEISV5.3.0.7; New Stack was included for BB-Fieldbus V1.1	New Stack, different behavior / parameter settings		-

5.2 Variant selection

The project does contains reference applications for all four fieldbusses, all shall be tested. With the firmwareupdate the fieldbus variant can be defined which is why we don't need different hardware and each tester gets one VHIP3 EVAL Board and one BB-Fieldbus EVAL Board.



5.3 Limitations and assumptions

The following assumptions have been considered for the test strategy:

- The requirements which describe the visible functionality and specification to the outside are complete.
- The system supplier Hilscher has adequately tested the SoC netX90 with respective fieldbus stack to meet the respective fieldbus specification. Conformance tests with the existing setup will be done by development.
- Developer tests on the hardware and software side are carried out and documented to ensure good basic quality.
- At the end of the development work, a full verification and validation is carried out in which all existing test cases are carried out. Depending on the extent of the errors found, either another full test or a regression test is performed to verify the error correction to ensure that the change is free of side effects. If the last official test cycle is a regression test, an impact analysis by the development department ensures that the unchecked areas are not affected by the change.

The following limitations have been considered for the test strategy:

- If the developed product has certain specifications or functionalities that are not specified in the requirements, then no explicit test cases are created for this.
- If requirements are not completely described, then the test may not be complete or it may also be faulty.
- The Hardware [EVAL Board VHIP 3 and EVAL Board BB-Fieldbus] is not under test.
Only the software functionalities of each reference application is to be tested.
- The tests will be only black box and reviews from documentation
- Although the official conformity tests are carried out, the interoperability cannot be guaranteed due to the huge number of all possible network topologies and different fieldbus devices.
- Since the tests are only carried out on an evaluation board of class VHIP3, no statement can be made as to whether the BB also works properly on boards which targets another VHIP classes (4, 5 and 6) or later derived products which build on those VHIP Platforms.
- With the reference application only the implemented functionality of the ifm fieldbus api can be tested.

5.4 Measures to achieve the test targets

- Official PN test bundle:
The official PN test environment, which is also used by the accredited laboratories when a PN device is to be officially certified, will be also used inhouse. Therefore, the evaluation board with the BB fieldbus PN and the reference application is tested here to determine whether it behaves according to the PN specification.
- Same for the Conformance Tests of EIP / ECAT / Modbus
- Functional test:
 - Of the reference application. Especially the behaviour which can be modified by the engineering tool.
- Fieldbus Tests will be carried out for Profinet with S7-1500series plcs, Ethernet/IP with AllenBradley CompactLogix PLCs EtherCAT with Beckhoff CX9020 PLCs. For Modbus the tests will be carried out with the official testtool "[Modbus Poll](#)".
- Interoperability:
The main focus is on ensuring that the fieldbus node works with controller of the market leader in the respective fieldbus without any problems. However, if there is time, controller from other manufacturers are used and principle tests will be carried out to determine if it works. Those tests be reduced to compatibility of the description file, configuration, validation of cyclic and if possible acyclic data.



5.5 Overview over planned testactivities

No	Technology / Measure	Target to achieve	DUT's / variant
1	Freetesting round	Experience / Curiosity driven testing to achieve testable first release version	For all fieldbus variants
2	Functional test (Ref.App. functionality to PN)	Proper functionality to Profinet shown	VH1000
2.1	Interoperability tests with Fanuc plcs	Proper basic functionality to Profinet shown	VH1000
3	Functional test (Ref.App. functionality to EIP)	Proper functionality to Ethernet/IP shown	VH1020
3.1	Interoperability tests with Omron/Mitsubishi plcs	Proper basic functionality to Ethernet/IP shown	VH1020
4	Functional test (Ref.App. functionality to ECAT)	Proper functionality to EtherCAT shown	VH1030
4.1	Interoperability tests with Omron plcs	Proper basic functionality to EtherCAT shown	VH1030
5	Functional test (Ref.App. functionality to MB)	Proper functionality to Modbus shown	VH1040

Table 5-2: Test strategy

6 Test documentation

6.1 Test plan (this document)

The test plan describes the test activities for one development phase according to the V-model of ecomatic. The test plan is part of test documentation.

6.2 Test specifications

The test specification is the collection of all test cases / test case descriptions. There may be (e.g. subject-specific) different test specifications and they will be documented in "Polarion". The relevant test specifications can be found [here](#).

6.2.1 Test case description

Test case descriptions have to be written in Polarion and can consists of a list (possibly hierarchically) of start situations, test actions and expected results as well as a statement, which of these actions should be executed to fulfill the testing strategy.

A typical test case description consists of at least the following:

- Scope of test which describes what shall be tested
- Which test parameters / test patterns will be used, how were they derived and what is the expected result
- Pre-conditions and hints for the test execution.
- Description of the test sequence / steps with expected result
- Links to further information/scripts/etc.

At least the following additional information is listed:

- State of the test case (draft, ready for review, valid, invalid and so on)
- Manual, semi-automated, automated test execution
- Test method (Black-Box, Performance, Interoperability, Review etc.)
- Test level (System, Product, etc.)
- Must be linked to the appropriate requirements



6.2.2 Test protocol

If an official test is planned based on one or more test specifications, one or more Polarion test runs with a unique name "ProjektNr-System-BB_Fieldbus-TRn-FieldbusType" are created in which the test results of a test case execution and any observations are documented.

The testrun content (testcase execution and testcase description) are regarded as the test protocol . Based on this, the test report will be created.

The individual test case description can consist of 1 to n test steps. For each step the tester must give a verdict and for the whole test case a overall verdict. The following verdicts are possible:

- **Pass:** Test step or complete test case has the verdict "Pass". The results matches with the expectation.
- **Fail:** Test step or complete test case has the verdict "Fail". The result does not match with the expectation. If possible the defect ID shall be added to the comment field.
- **Not planned:** In test steps this verdict is used for steps that are not applicable. In such a case the test case shall be revised. Complete test cases will have this verdict if run in a wrong test run. In such cases comment shall be added to the comment field.
- **Blocked:** Test step or complete test case could not be done (e.g. due to missing functionality). In such cases comment shall be added to the comment field.

6.3 Reports

The results of the test runs will be documented in test reports. The individual reports must be prepared close to finalizing the test activities.

6.3.1 Test reports

The results of the verification activities will be documented in one or several test reports.

The following test report will be created and stored in the SVN directory according QM-plan:

- 154291 Testreport VHIP BB Fieldbus

The test report summarizes the information on all test protocols which are required to achieve the necessary test objective (e.g. full test, regression test, etc.) for document the test object.

In order to ensure the traceability of bug fixes and their retests that were carried out during the official testing activities, this information is also included in the report

The test report must contain the test sequences that were performed for the specified test objective. Earlier test runs and tests accompanying development are not part of the "final" test report.

7 Test resources

7.1 DUT type, amount and configurations

In order to carry out the approval activities the following test equipment is required (see chapter "Test object").

Article no.	Description	Version type	Amount
EVAL Boards VHIP3 STM32F777 EVAL Boards BB-Fieldbus			4
			Sum overall: 4



The detailed information about the used test equipment will be documented in the test run and/or the test report (serial number or similar identifier).

7.2 Effort

7.2.1 Tasks to be done prior to testing

- Develop functionalities to automate BB-Fieldbus tests (for each fieldbus)

7.2.1.1 Preparation of test setups

No setups so far.

7.2.1.2 Creation of test specifications

Since the ifm ecomatic already has devices with each fieldbus, an attempt is made to take over generally valid test cases from there. However, within the scope of the BB Fieldbus product, there will also be many test cases that have to be newly created.

If it makes economic sense, we try to automate test cases in particular that can be carried out during development.

Documentation of the tests will be inside of the test case.

Project and function blocks necessary to run the tests will be saved in SVN.

7.2.1.3 Further test planning, if needed

Not planned.

7.2.2 Test execution

The necessary testing effort is divided into the following phases:

- Creation of the test cases
- Execution of the test cases
- Reporting

The effort required to create test cases depends on the requirements to be checked. Many complex requirements lead to many test cases, while simple ones lead to a small number of test cases.

The effort required to carry out the test cases depends on the following factors:

- Number of test cases to be carried out
- Complexity of the test cases
- Number of executions due to bug fixes and their retests

The effort for creating the test report is rather low in relation to creating / executing the test cases, but it can still take a few days. Overall, a concrete estimate of the effort is not useful here. It is carried out as part of the project plan and updated cyclically. Typically the testing is planned to be sequenced in 4 steps:

1. Free testing / sandbox test round (~5 days)
Testing heavily relaying on experience / curiosity of the tester, functionalities shall be used / played around to find new test scenarios; to get a feeling of the DuT. Old (postphoned) CR can be retested with the new version and new test cases created / requested.
2. First test round of the existing test cases to validate the new version against the old stand.
3. Bug fixes and retesting of the bug fixes. Tests will be handled inside of the Change Request and do not require a



own test round.

4. Final test round with all test cases to prove no other working functionality was broken with the fixes.

7.3 Tester

Testing is planned to be done by the following persons:

Role / Task	Assignees
Test engineer functional tests	Alexander Meier (ECAT) Hajer Mouldi (PN and EIP) John Wilkinson (Modbus)
Test manager / Test engineer functional tests	Bastian Ruess (Testmanagement)

7.4 Project schedule

The responsibility for creation of a master schedule for the activities derived from this test plan is held by the project management and is part of the project plan, the during the project updated working tasks are defined here: [154291_Arbeitspakete.xlsx](#)

7.5 Test environment

The test environment describes the facilities with the necessary equipment, tools, aids, etc. necessary to perform the tests in order to fulfill the tasks of the test plan and the related test specifications.

7.5.1 Test automation

Automation of the test shall be implemented in multiple steps, depending on the availability of the resources t:

1. Testcases shall be grouped by their option to automate, using the automation enumeration of the test cases.
Future test cases shall be written with an focus to limit the manual tests.
2. Automate the plc functionalities to gain a partly automated functionality. Tester will be required to start the test and gather the results, or make entries in a parameter table.
3. Fully automate the test case by providing scripts to upload configuration and project / function blocks to plc.
4. Develop automatic procedure to test a test run in polarion without manual labor.

7.5.2 Software / Tools

The necessary software tools such as Siemens TIA Portal / Studio 5000 / TwinCAT and other standard software are available in the company at the moment and do not have to be listed here. The used tools will be listed later either in the testcase if a specific version is required or are stated in the comment of the testcase as part of the testsetup.

For Modbus Licenses for the Tool Modbus Poll were acquired.

7.5.3 Hardware / test equipment

The necessary hardware and test equipment especially the official conformance test environment are defined by the respective specifications which can be found in [4]...[6].

For the testing of the Profinet Fieldbus communication, the Siemens PLCs of the series 1500 has been proven to be successful in previous product tests so they are used for the testing of the BB also.

For the testing of the Ethernet/IP Fieldbus communication, the AllenBradley PLCs of the CompactLogix series e.g. 1769 has been proven to be successful in previous product tests so they are used for the testing of the BB also. There is a more powerful PLC series with the ControlLogix but those weren't necessary until now.

For the testing of the EtherCAT communication a Beckhoff PLC will be used since they are by far the biggest market



particent.

Other laboratory equipment like PSU'S, Multimeters etc. are available in the company at the moment and do not have to be listed here. The used hardware equipment will be listed later by the tester in the comment of the testcase of the performed Testrun.

8 Assessment of test activities

Evaluation of the results of the verification and validation activities.

8.1 Assessment of test activities

8.1.1 Completeness

The evaluation of the testing activities includes an assessment of the completeness of the tests. Ideally, there is at least one test case per requirement.

8.1.2 Condition of the test specification

The quality of the test activities also includes the state of the test specification.

All test case descriptions must be reviewed and approved prior to start of testing. If the need for test case changes, their status shall be set back to Ready_for_Review when modified and a re-review shall take place (and the test specification shall also be approved again).

8.1.3 Test environment

The quality of the used test setups often depends on whether they have to be newly created for a specific test purpose or whether it is possible to fall back on existing setups that have proven their reliability and meaningfulness.

In the specific case, the official test setup for the PN conformance test in particular has a very high information value, as this is elementary in order to have the device officially certified. However, this structure has also been rebuilt as part of this project and must be learned by the supervisors.

At least there is the test setup for the long term test which is used for other ifm ecomatic PN devices. This test setup has already proven in the past that it can detect errors in the device.

8.2 Assessment of the test results

8.2.1 Pass/fail criteria

Ideally, all test cases should be passed. In the event of a failed test case, the TM or the whole team of the project can decide to accept the wrong behaviour. However, this can only be the case for misconduct that would not prevent the official certification of the device. If there are any a document shall be created e.g. known bug List or Releasenotes to inform about the state.

Remark: The "pass" and "fail" criteria are specified for the individual test case.

8.3 Test planning risks

The following risks could affect or influence the test plan:

- Only few internal resources available for this project.
- Employees who are testing EtherCAT / Modbus devices for the first time
- Test cases have to be written without possibility to run those on a DUT in parallel. This has to be done later, which makes revised versions very likely (-> more effort)
- Limited test equipment due to cost constraints possible
- Test cases are not completed on time
- Tight schedule, no time for an intensive testing (e.g. Jitter, Cycletimes, Longterm effects)



- Adjustments for new GSDML / Firmware in Testcases / Testsetups can be time consuming.
- Reference Application defines the possible testable feature set.

Therefore not every function of the fieldbus API is tested right now.